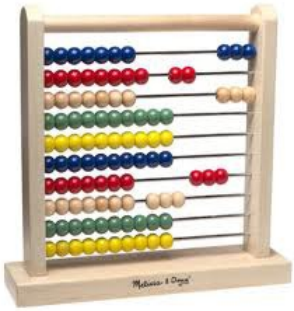
Early Computing: Crash Course Computer Science #1

Study online at https://quizlet.com/_5ytw7o

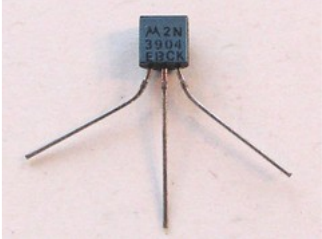
1. **The earliest recognized device for computing was the _____, invented in Mesopotamia around 2500 BCE.**

   abacus

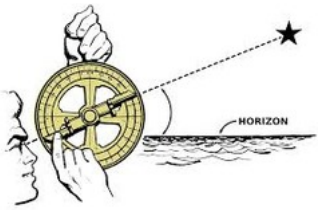
2. **Transistors in just your smartphones, computers can seem pretty complicated, but really,**

   they're just simple machines that perform complex actions through many layers of abstraction

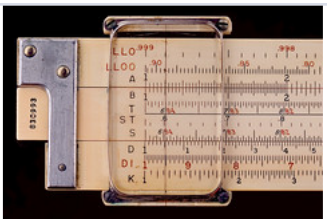
3. **astrolabe**

   An instrument used by sailors to determine their location by observing the position of the stars and planets

4. **slide rule**

   A manual calculator invented by William Oughtred that uses John Napier's logarithms to perform complex engineering and scientific calculations

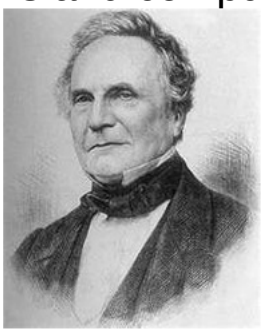


5. **These devices made something that was previously laborious to calculate much faster, easier, and often more accurate -- it lowered the barrier to entry,**

and at the same time, amplified our mental abilities

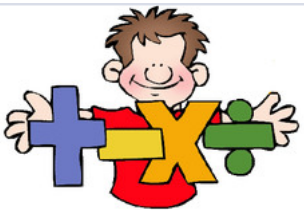
6. **Charles Babbage**

He is known as the "Father of Computers." He developed the Difference Engine and the Analytical Engine. These two machines were powered by steam to move the punch cards and operate the gears. His ideas were later used to design calculators and computers.



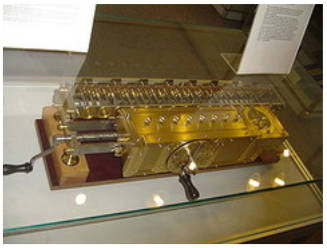
7. **In the olden days a computer was a person who**

did calculations, sometimes with the help of machines, but often not.



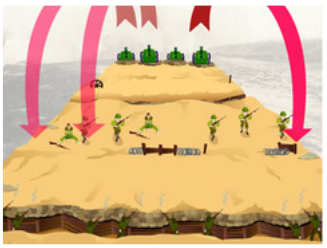
8. **Step Reckoner**

a digital mechanical calculator invented by the German mathematician Gottfried Wilhelm Leibniz



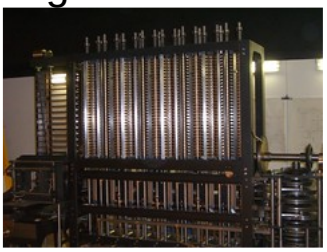
9. **The military was among the first to apply**

computing to complex problems.



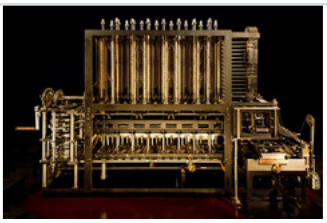
10. **Difference Engine**

automatic mechanical calculator designed to tabulate polynomial functions created by Charles Babbage



11. **Analytical Engine**

1st mechanical computer. Ran off of punched cards. Created by Charles Babbage, but never completed by him.



12. **Ada Lovelace**

She is considered to be the first computer programmer. She wrote a computer language for the Analytical Engine.

13. **Herman Hollerith**

He invented the "Tabulating Machine." The Tabulating Machine used punch cards to store information. It was later used to tabulate the U.S. Census.

14. **The Tabulating Machine Company, which later merged with other machine makers in 1924 to become**

The International Business Machines Corporation or IBM

Electronic Computing: Crash Course Computer Science #2

Study online at https://quizlet.com/_5ytx0s

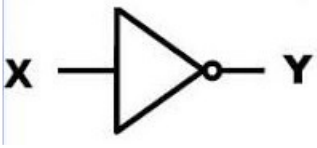
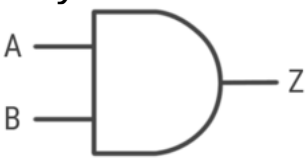
1.	**One of the largest electro-mechanical computers built was the _____, completed in 1944 by IBM for the Allies during World War 2. It contained 765,000 components, three million connections, and five hundred miles of wire. To keep its internal mechanics synchronized, it used a 50-foot shaft running right through the machine driven by a five horsepower motor.**	Harvard Mark I
2.	**In a relay, there is a**	control wire that determines whether a circuit is opened or closed. The control wire connects to a coil of wire inside the relay. When current flows through the coil, an electromagnetic field is created, which in turn, attracts a metal arm inside the relay, snapping it shut and completing the circuit.
3.	**The Harvard Mark I could do**	3 additions or subtractions per second; multiplications took 6 seconds, and divisions took 15. And more complex operations, like a trigonometric function, could take over a minute.
4.	**In September 1947, operators on the Harvard Mark II pulled a dead moth from a malfunctioning relay.**	"From then on, when anything went wrong with a computer, we said it had bugs in it." And that's where we get the term computer bug.

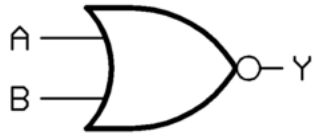
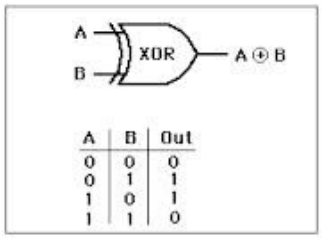
5.	**John Ambrose Fleming**	In 1904, English physicist who developed a new electrical component called a thermionic valve, which housed two electrodes inside an airtight glass bulb - this was the first vacuum tube.
6.	**thermionic emission**	the release of electrons from the tungsten filament when the electrical current passes through it and heats the filament
7.	**diode**	a semiconductor device with two terminals, typically allowing the flow of current in one direction only.
8.	**triode**	a vacuum tube that has three electrodes
9.	**Colossus Mk 1**	first large-scale use of vacuum tubes for computing that was designed by engineer Tommy Flowers and completed in December of 1943. It was installed at Bletchley Park, in the UK, and helped to decrypt Nazi communications.
10.	**Alan Turing**	created first operational electronic digital computer in the 1940s and turing machine
11.	**ENIAC**	Electronic Numerical Integrator and Comput-

	er; U.S. government-sponsored machine developed to calculate the settings used for weapons
12. **In 1947, Bell Laboratory scientists John Bardeen, Walter Brattain, and William Shockley invented the _____.**	transistor
13. **What made the transistor amazing?**	-solid state component reducing wear and tear - it could switch between on and off states 10,000 times per second -could be made smaller than the smallest possible relays or vacuum tubes
14. **IBM 608**	Released in 1957 - the first fully transistor-powered, commercially-available computer. It contained 3,000 transistors and could perform 4,500 additions, or roughly 80 multiplications or divisions, every second.

Boolean Logic & Logic Gates: Crash Course Computer Science #3

Study online at https://quizlet.com/_5ytxpr

1.	**Binary**	A way of representing information using only two options.
2.	**ternary**	three part form (ABA)
3.	**quinary**	consisting of five
4.	**The more intermediate states there are,**	the harder it is to keep them all separate.
5.	**Boolean Algebra**	A mathematical notation for expressing two-valued logical functions. Invented by George Boole, a self-taught English mathematician in the 1800s. He was interested in representing logical statements that went "under, over, and beyond"
6.	**There are three fundamental operations in Boolean Algebra:**	a NOT, an AND, and an OR operation.
7.	**In boolean terms, when the input is true,**	the output is true.
8.	**In boolean terms, when the input is false,**	the output is false.
9.	**NOT gate**	input will be off and the output will be on 
10.	**AND gate**	takes two inputs, but still has a single output. In this case the output is only true if both inputs are true. 

11. **OR gate**	where only one input of the two inputs has to be true for the output to be true 
12. **Exclusive OR (XOR)**	where only one input of the two inputs has to be true for the output to be true or if both inputs are true, the output is false. 
13. **Logic Gate**	An electrical circuit that produces output as determined by the corresponding logic table.

A	B	Out
0	0	0
0	1	1
1	0	1
1	1	0

Representing Numbers and Letters with Binary: Crash Course Computer Science #4

Study online at https://quizlet.com/_5ytyws

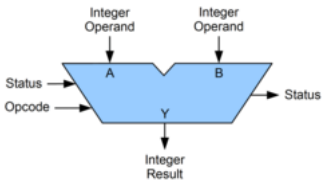
1.	**Instead of 'true' and 'false',**	we can call these two states 'one' and 'zero'
2.	**base 2**	The name given to the counting system for Binary
3.	**binary number 101**	5
4.	**number in binary: 10110111**	1*128 + 0*64 + 1*32 + 1*16 + 0*8 + 1*4 + 1*2 + 1*1 = 183
5.	**11 in binary**	3
6.	**11001010 in binary**	202
7.	**Each of these binary digits 1 or 0 is called a**	bit
8.	**8-bit numbers with their lowest value of**	0 and highest value of 255, which requires all eight bits to be set to 1. That's 256 different values, or 2^8
9.	**A byte is**	8 bits
10.	**Terabyte (TB)**	eight trillion 1s and 0s or said another way 1 trillion bytes
11.	**Kilobyte (KB)**	2^10 bytes, or 1024
12.	**Megabyte (MB)**	1 million bytes or 8 million bits
13.	**Gigabyte (GB)**	1 billion bytes or 8 billion bits
14.	**64-bit computer**	operate in chunks of 64 bits
15.	**The largest value a 64-bit number can represent is around**	9.2 quintillion

16.	**Format used by IEEE 754 Standard**	[significand] x 10^[exp]
17.	**IEEE 754 Standard**	defines formats for 16, 32, 64, and 128 bit floating point arithmetic with a binary exponent as well as 64 and 128 with a decimal exponent
18.	**32-bit floating-point number,**	the first bit is used for the sign of the number, positive or negative. The next eight bits are used to store the exponent, and the remaining 23 bits are used to store the significand.
19.	**ASCII (American Standard Code for Information Interchange)**	a code for representing English characters as numbers, with each letter assigned a number from 0 to 127
20.	**A lowercase "a" is represented by the number**	97
21.	**A capital "A"**	65
22.	**interoperability**	The ability, generally by adoption of standards, of systems to work together
23.	**mojibake**	'scrambled text'
24.	**Unicode**	A character code that enables most of the languages of the world to be symbolized with a special character identification.

How Computers Calculate - the ALU: Crash Course Computer Science #5

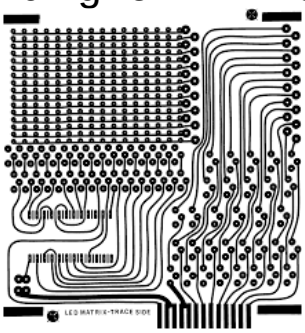
Study online at https://quizlet.com/_5yw54n

1.	**ALU**	Arithmetic and Logic Unit - does all mathematical calculations and makes all logical decisions
2.	**Intel 74181**	the first complete ALU that fit entirely inside a single chip, which was a huge engineering feat at the time
3.	**There are only four possible input combinations. The first three are:**	0 + 0 = 0 1 + 0 = 1 0 + 1 = 1
4.	**half adder**	Logic circuit with two inputs and two outputs. The inputs are a bit from the augend and a bit from the addend. The outputs are the sum bit produced by the addition of the bit from the addend with the bit from the augend and the resulting carry (COUT) bit, which will be added to the next stage.
5.	**CARRY bit**	Bit used to indicate when an carry or borrow has been generated in an arithmetic operation
6.	**SUM bit**	in one column is a two-bit (two-digit) number, the least significant figure is written as part of the total sum and the most significant figure is "carried" to the next left column.
7.	**Full Adder**	A circuit that computes the sum of two bits, taking an input carry bit into account
8.	**8-bit ripple carry adder**	a circuit that takes two 8-bit numbers, let's call them A and B, and adds them together. Let's start with the very first bit of A and B, which we'll call A0 and B0. At this point, there is no carry bit to deal with, because this is our first addition. So we can use our half adder to add those two bits together. The output is SUM0. Now we want to add A1 and B1 together. It's possible there was a carry from the previous addition of A0 and B0, so this time we need to use a full adder that also inputs the carry bit. We output this result as SUM1. Then, we take any carry from this full adder and run it into the next full adder that handles A2

		and B2. And we just keep doing this in a big chain so all 8 bits have been added. Notice how the carry bits ripple forward to each subsequent adder.
9.	**Notice how our last full adder has a carry out. If there is a carry into the ninth bit, it means the sum of the two numbers is too large to fit into eight bits. This is called an _____.**	overflow.
10.	**Overflow**	can usually cause errors and unexpected behavior
11.	**What was the greatest right of passage for Pac-Man players?**	To make it past level 255, the largest number storable in 8 bits, to level 256, the ALU overflowed. This caused a bunch of errors and glitches, making the level unbeatable.
12.	**carry-look-ahead adder**	Faster than the 8-bit ripple carry adder , but ultimately does exactly the same thing: adds binary numbers.
13.	**Special symbol to wrap up the complexity of an 8-bit ALU?**	

Registers and RAM: Crash Course Computer Science #6

Study online at https://quizlet.com/_5ywd9p

1.	**RAM**	Random Access Memory - temporary place to store material that works quickly. erased when computer turns off. Volatile.
2.	**AND OR Latch**	It has two inputs, a "set" input, which sets the output to a 1, and a "reset" input, which resets the output to a 0. If set and reset are both 0, the circuit just outputs whatever was last put in it. In other words, it remembers a single bit of information! Memory!
3.	**The action of putting data into memory is called _____.**	writing
4.	**The action of getting the data out is called _____.**	reading
5.	**Gated Latch**	can be opened or closed
6.	**If we put 8 latches side-by-side, we can store 8 bits of information like an 8-bit number. A group of latches operating like this is called**	a register, which holds a single number, and the number of bits in a register is called its width.
7.	**16x16 latch matrix**	For 256 bits, we need a 16 by 16 grid of latches with 16 rows and columns of wires. To activate any one latch, we must turn on the corresponding row AND column wire. 
8.	**For 256 bits of memory, we only need**	35 wires - 1 data wire, 1 write enable wire, 1 read enable wire, and 16 rows and columns for the selection

9.	**Multiplexer**	A circuit that directs one of several digital signals to a single output, depending on the states of several select inputs.
10.	**abstraction**	the quality of dealing with ideas rather than events
11.	**uniform bank of addressable memory**	256 addresses, and at each address, we can read or write an 8-bit value
12.	**To address a gigabyte - or a billion bytes of memory -**	we need 32-bit addresses.
13.	**If we opened up a stick of 1 megabyte RAM you would see is**	8 memory modules soldered onto the board. If we carefully opened up one of these modules and zoomed in, you would see are 32 squares of memory. Zoom into one of those squares, and we can see each one is comprised of 4 smaller blocks. If we zoom in again, we get down to the matrix of individual bits. This is a matrix of 128 by 64 bits. That's 8192 bits in total. Each of our 32 squares has 4 matrices, so that's 32 thousand, 7 hundred and 68 bits. And there are 32 squares in total. So all in all, that's roughly 1 million bits of memory in each chip. Our RAM stick has 8 of these chips, so in total, this RAM can store 8 million bits, otherwise known as 1 megabyte.
14.	**SRAM**	static random access memory
15.	**DRAM**	dynamic random access memory
16.	**NVRAM**	Non-Volatile Random Access Memory
17.	**Flash Memory**	Type of nonvolatile memory that can be erased electronically and rewritten.

The Central Processing Unit (CPU): Crash Course Computer Science #7

Study online at https://quizlet.com/_5ywgnf

1.	**CPU**	Central Processing Unit whose job is to execute programs.
2.	**Programs**	are made up of a series of individual operations called instructions, because they instruct the computer what to do.
3.	**micro-architecture**	a description of the electrical circuitry of a computer, central processing unit, or digital signal processor that is sufficient for completely describing the operation of the hardware.
4.	**Two types of computer memory**	Registers and RAM
5.	**We can assign an _____ to each instruction supported by our CPU.**	ID
6.	**operation code**	Opcode. First part of machine language instruction. Specifies which operation needs to be carried out (e.g., addition, store)
7.	**Instruction Address Register**	stores the memory address of the current instruction
8.	**Fetch phase**	First phase of a CPU's operation where we retrieve our first instruction. First, we wire our Instruction Address Register to our RAM module. The register's value is 0, so the RAM returns whatever value

		is stored in address 0. Then this value is copied into our instruction register.
9.	**Decode phase**	Second phase where we figure out what that instruction is so we can execute it.
10.	**In this case the op-code, which is the first four bits, is: 0010.**	This op-code corresponds to the "LOAD A" instruction, which loads a value from RAM into Register A. The RAM address is the last four bits of our instruction which are 1110, or 14 in decimal.
11.	**Control Unit (CU)**	Processor component that manages the execution of instructions during the fetch-execute cycle.
12.	**Execute Phase**	Last phase where the CU completes the instructions.
13.	**Clock Speed**	The speed, or frequency, expressed in MHz, that controls activity on the motherboard and is generated by a crystal or oscillator located somewhere on the motherboard.
14.	**The very first, single-chip CPU was the Intel 4004, a 4-bit CPU released in 1971 had a mind-blowing clock speed of**	740 Kilohertz -- that's 740 thousand cycles per second.
15.	**overclocking a computer**	when you modify the clock to speed up the tempo of the CPU
16.		

Chip makers often design CPUs with enough tolerance to handle a little bit of overclocking, but too much	can either overheat the CPU or produce gobbledygook as the signals fall behind the clock
17. **underclocking**	Making your CPU run slower so you can save a lot of power, which is important for computers that run on batteries
18. **dynamic frequency scaling**	processors can increase or decrease their clock speed based on demand

Instructions & Programs: Crash Course Computer Science #8

Study online at https://quizlet.com/_5z1dh9

1.	**To make a basic, but functional Central Processing Unit - or CPU - you need:**	ALU, control unit, some memory, and a clock
2.	**software**	(computer science) written programs or procedures or rules and associated documentation pertaining to the operation of a computer system and that are stored in read/write memory
3.	**hardware**	The machines, wiring, and other physical components of a computer or other electronic system
4.	**JUMP**	This causes the program to "jump" to a new location. This is useful if we want to change the order of instructions, or choose to skip some instructions.
5.	**JUMP_NEGATIVE**	only jumps the program if the ALU's negative flag is set to true
6.	**HALT**	tells the computer when to stop processing
7.	**STORE**	saves data into memory location
8.	**Infinite loops**	keep repeating until the program is interrupted
9.	**conditional jumps**	JUMP IF EQUAL and JUMP IF GREATER.
10.	**Software can do something our hardware could not. The ALU didn't have the functionality**	to divide two numbers, instead it's the program we made that gave us that functionality.
11.	**Immediate Value**	When the CPU sees an instruction that needs something like a JUMP instruction, it knows it must also fetch the ad-

	dress to jump to, which is saved immediately behind the JUMP instruction in memory.
12. **A modern computer processor, like an Intel Core i7,**	has thousands of different instructions and instruction variants, ranging from one to fifteen bytes long.
13. **1971, Intel released the 4004 processor that supported**	46 instructions, shown here. Which was enough to build an entire working computer.

Advanced CPU Designs: Crash Course Computer Science #9

Study online at https://quizlet.com/_5z1gbs

1.	**Gigahertz speeds**	billions of instructions executed every second
2.	**Modern computer processors now have special circuits for things like**	graphics operations, decoding compressed video, and encrypting files - all of which are operations that would take many many many clock cycles to perform with standard operations.
3.	**bus**	data has to be transmitted to and from RAM along sets of data wires
4.	**cache**	a little piece of RAM right on the CPU
5.	**cache dirty bit**	a special flag for each block of memory it stores
6.	**When the cache is full, but a new block of memory is being requested by the processor. Before the cache erases the old block to free up space,**	it checks its dirty bit, and if it's dirty, the old block of data is written back to RAM before loading in the new block.
7.	**Instruction pipelining is a processing technique**	that allows the processor to operate on multiple stages of instruction at a time with the pipeline keeping the instructions in order.
8.	**parallelize**	While one instruction is getting executed, the next instruction could be getting decoded, and the instruction beyond that fetched from memory. All of these separate processes can overlap so that all parts of the

		CPU are active at any given time.
9.	**Pipelined processors have to look ahead for**	data dependencies
10.	**out-of-order execution**	dynamically reorder instructions with dependencies in order to minimize stalls and keep the pipeline moving
11.	**A simple pipelined processor will perform a _____ when it sees a jump instruction, waiting for the value to be finalized.**	long stall
12.	**speculative execution**	the ability of some CPU's to execute instructions ahead of the current location and beyond a conditional branch. The results of these instructions will be usable only if the CPU guessed the branch direction correctly.
13.	**branch prediction**	An approach to dealing with conditional BRANCHes in which the CPU guesses whether a branch condition will be true or false based on past experience; a form of parallel processing.
14.	**multi-core processors**	Contains two or more which two or more processors have been attached for enhanced performance, reduced power consumption, and more efficient simultaneous processing of multiple tasks.

15. **When this video was made, the world's fastest computer was**	located in The National Supercomputing Center in Wuxi, China. The Sunway TaihuLight contains a brain-melting 40,960 CPUs, each with 256 cores
16. **FLOPS**	The number of floating point operations per second that a computational device can perform. A typical measure of the power of a device to perform calculations.
17. **The Sunway TaihuLight can process**	93 Quadrillion -- that's 93 million-billions -- floating point math operations per second
18. **Crysis**	a first-person shooter video game series developed by German developer Crytek and published by Electronic Arts

Early Programming: Crash Course Computer Science #10

Study online at https://quizlet.com/_5z1jmr

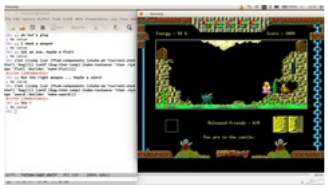
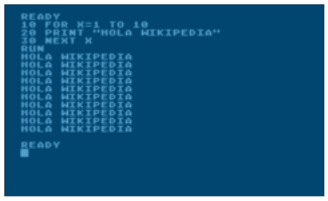
1.	**Many consider Jacquard's loom to be one of the earliest forms of**	programming.
2.	**plugboards**	A control panel full of little sockets, into which a programmer would plug cables to pass values and signals between different parts of the machine.
3.	**Von Neumann Architecture**	Traditional computer architecture that forms the basis of most digital computer systems. A single control unit manages program control flow following a linear sequence of fetch-decode-execute
4.	**62,500 punch cards**	equivalent to roughly 5mb of data
5.	**Panel programming**	used huge panels full of switches and buttons, and there were indicator lights to display the status of various functions of values and memories
6.	**The first commercially successful home computer was the ALTAIR 8800 were sold in two versions:**	preassembled and as a kit.
7.	**To program the 8800, you'd literally**	toggle the switches on the front panel to enter the binary opcodes and instructions you wanted. Then, you'd press the deposit button to write that value into memory. Then, in the next location in memory, you'd toggle the switches again for your next instruction, deposit it, and so on.

The First Programming Languages: Crash Course Computer Science #11

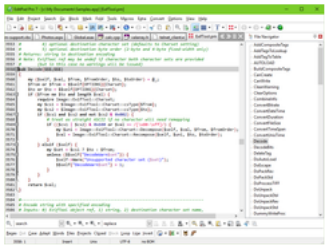
Study online at https://quizlet.com/_5z53r5

1.	**Programming at the hardware level is cumbersome and inflexible, so programmers wanted a more versatile way to program computers -**	what you might call a "softer" medium.
2.	**Machine Language**	The language made up of binary-coded instructions that is used directly by the computer
3.	**Pseudo-Code**	a notation resembling a simplified programming language, used in program design.
4.	**mnemonics**	memory aids, especially those techniques that use vivid imagery and organizational devices
5.	**Assembler**	A program that translates a symbolic version of instructions into the binary version.
6.	**Dr. Grace Hopper**	Mathematician who worked on developing software for computers; developed the Compiler; and created FORTRAN; coined the phrase "debug"
7.	**They "patched" some bugs in this program by literally**	putting patches of paper over the holes on the punch tape.
8.	**A-0**	high-level programming language called "Arithmetic Language Version 0"
9.	**Compiler**	A computer program created to read an entire program and convert it into a lower-level language and ultimately to assembly language used by the processor.
10.	**Python**	

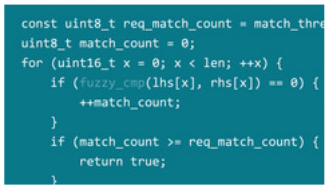
		an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991
11.	**variables**	abstractions for needed memory locations
12.	**FORTRAN**	One of the original high level languages, short for Formula Translation, created by John Backus to make programming easier for math and science applications.
13.	**John Backus, the FORTRAN project director, said:**	"Much of my work has come from being lazy."
14.	**Programs written in FORTRAN were _____ than equivalent handwritten assembly code**	20 times shorter
15.	**Committee on Data Systems Languages**	a consortium formed in 1959 to guide the development of a standard programming language that could be used on many computers
16.	**COBOL**	An acronym for common business-oriented language, a compiled English-like computer programming language designed for business use

17.	**This notion is called write once,**	run anywhere
18.	**ALGOL**	ALGOrithmic Language released in 1960 - Provided a standard notation for computer scientists to publish algorithms in journals - Included structured control statements for sequencing (begin-end) blocks, loops (for loop), and selection (if and if-else statements) - Supported different numeric types - Introduced the array structure - Supported procedures, including recursive procedures
19.	**LISP**	a family of computer programming languages with a long history and a distinctive, fully parenthesized prefix notation. Originally specified in 1958, it is the second-oldest high-level programming language in widespread use today 
20.	**BASIC**	an acronym for Beginner's All-purpose Symbolic Instruction Code) is a family of general-purpose, high-level programming languages whose design philosophy emphasizes ease of use. 
21.	**Pascal**	an imperative and procedural programming language, which Niklaus Wirth designed in 1968-69 and published in 1970, as a small, efficient language intended to encourage good programming practices using structured programming and data structuring. It is named in honor of the French mathematician, philosopher and physicist Blaise Pascal.

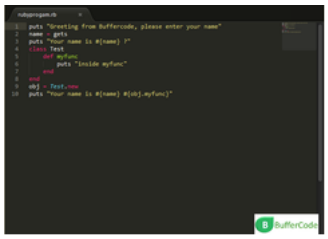
22. **PERL** a family of two high-level, general-purpose, interpreted, dynamic programming languages, Perl 5 and Perl 6. Though Perl is not officially an acronym, there are various backronyms in use, including "Practical Extraction and Reporting Language"



23. **C++** a statically-typed, free-form, (usually) compiled, multi-paradigm, intermediate-level general-purpose middle-level programming language." In simple terms, C++ is a sophisticated, efficient and a general-purpose programming language based on C. It was developed by Bjarne Stroustrup in 1979.

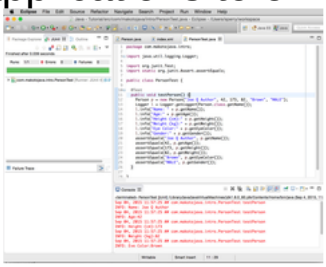


24. **ruby** a dynamic, interpreted, reflective, object-oriented, general-purpose programming language. It was designed and developed in the mid-1990s by Yukihiro "Matz" Matsumoto in Japan.



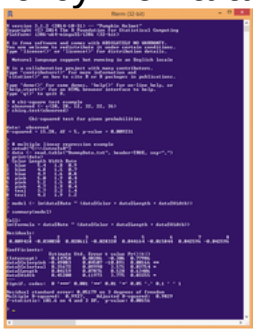
25. **Java** An object-oriented programming language designed specifically for programs (particularly multimedia) to be

used over the Internet. Java allows programmers to create small programs or applications to enhance websites.



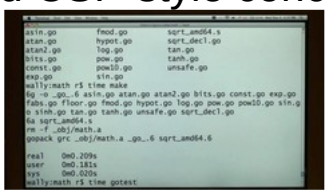
26. **C#**

a general-purpose, multi-paradigm programming language encompassing strong typing, imperative, declarative, functional, generic, object-oriented, and component-oriented programming disciplines. It was developed around 2000 by Microsoft within its .NET initiative and later approved as a standard by Ecma and ISO.



27. **Go**

a statically typed, compiled programming language designed at Google by Robert Griesemer, Rob Pike, and Ken Thompson. Go is syntactically similar to C, but with the added benefits of memory safety, garbage collection, structural typing, and CSP-style concurrency.



28. **Swift**

a general-purpose, multi-paradigm, compiled programming language developed by Apple Inc. for iOS, macOS, watchOS, tvOS, Linux and z/OS. Swift is designed to work with Apple's Cocoa and Cocoa Touch frameworks and the large body of existing Objective-C code written for Apple products.

29. **It's extremely likely that the web browser you're using right now was written in**	C++ or Objective-C.
30. **Many consider the holy grail of programming to be the use of**	"plain ol' English", where you can literally just speak what you want the computer to do, it figures it out, and executes it.

Programming Basics: Statements & Functions: Crash Course Computer Science #12

Study online at https://quizlet.com/_5z6381

1.	**syntax**	The set of rules that govern the structure and composition of statements in a language.
2.	**assignment statement**	an instruction that assigns a value to something, such as to the property of an object
3.	**A program, which is a list of instructions, is a bit like a _____**	recipe
4.	**initialize our variables**	set their initial value
5.	**Control Flow Statements**	Decision making structures have one or more conditions to be evaluated or tested by the program.
6.	**If**	A statement consists of a boolean expression followed by one or more statements.
7.	**If Else**	An _____ statement can be followed by an optional _____ statement, which executes when the boolean expression is false.
8.	**Nested If**	You can use one if or else if statement inside another if or else if statement(s).
9.	**Switch**	A _____ statement allows a variable to be tested for equality against a list of values.
10.	**Conditional Statements**	Operations that ask a question and select the next instruction to carry out based on the answer to that question.
11.	**While Loop**	a programming construct used to repeat a set of commands (loop) as long as (while) a boolean condition is true

12. **For Loop**	A particular kind of looping construct provided in many languages. Typically, a for loop defines a counting variable that is checked and incremented on each iteration in order to loop a specific number of times.
13. **subroutines**	in problem solving, specific procedures for solving familiar, well-defined problems
14. **functions**	programming languages can package pieces of code
15. **RETURN statement**	A statement of the form return <expression> It is placed at the end of a fruitful function to return a value
16. **Libraries**	huge bundles of pre-written functions

Intro to Algorithms: Crash Course Computer Science #13

Study online at https://quizlet.com/_5z69cr

1.	**algorithm**	a methodical, logical rule or procedure that guarantees solving a particular problem. .
2.	**Some algorithms are better than others even if they produce equal results. Determiners can be**	the fewer steps it takes to compute or how much memory it uses.
3.	**Computer Science**	the study of the principles and use of computers.
4.	**array**	a series of items
5.	**Selection Sort**	A sort algorithm that repeatedly scans for the smallest item in the list and swaps it with the element at the current index. The index is then incremented, and the process repeats until the last two elements are sorted.
6.	**Big O Notation**	a notation that expresses computing time (complexity) as the term in a function that increases most rapidly relative to the size of a problem
7.	**Merge Sort**	A list is split into individual lists, these are then combined (2 lists at a time).
8.	**Graph Search**	away of mapping all of the data we give the platform together in a really useful way; best example of 'Social Search' - the premise of creating a search engine based not on websites but on entities - people, places and things
9.	**Dijkstra's algorithm**	an optimal greedy algorithm to find the minimum distance and shortest path in a weighted graph from a give start node.
10.	**Bubble Sort**	Moving through a list repeatedly, swapping elements that are in the wrong order.
11.	**Spaghetti Sort**	

a linear-time, analog algorithm for sorting a sequence of items, introduced by Alexander Dewdney. It sorts a sequence of items requiring O stack space in a stable manner. It requires a parallel processor.

Data Structures: Crash Course Computer Science #14

Study online at https://quizlet.com/_606vip

1.	**Data Structures**	Variables that store data for direct or indirect processing used in computer programs.
2.	**Arrays**	a collection of items stored in a contiguous memory location and addressed using one or more indices
3.	**strings**	arrays of characters, like letters, numbers, punctuation and other written symbols
4.	**null character**	denotes the end of the string in memory
5.	**string concatenation function, or "string cat,"**	takes in two strings, and copies the second one to the end of the first
6.	**matrix**	an array of arrays
7.	**a=j[2][1]**	tells the computer you're looking for the item in subarray 2 at position 1
8.	**a=j[2][0][18][13][3]**	5th dimensional matrix
9.	**struct**	Groups of variables bundled together
10.	**node**	stores a variable, like a number, and also a pointer
11.	**pointer**	a special variable that points to a location in memory
12.	**linked list**	a flexible data structure that can store many nodes; can be re-ordered, trimmed, split, reversed
13.	**queue**	goes in order of arrival

14. **stack**	goes in order of last data pushed in
15. **FIFO**	First-In First-Out
16. **Last-In First-Out**	LIFO
17. **Enqueue**	operation adds an item to a queue
18. **Dequeue**	operation removes an item from the queue
19. **Pushing and Popping**	official terms used to describe adding and completing tasks in a stack
20. **binary tree**	a tree in which each node has at most two children
21. **The top most node is called the _____. And any nodes that hang from other nodes are called children nodes.**	root
22. **The top most node is called the root. And any nodes that hang from other nodes are called _____ nodes.**	children
23. **Nodes above children are called _____ nodes.**	parent
24. **Leaf Nodes**	any nodes that have no children
25. **important property of trees**	there's a one-way path from roots to leaves
26. **Tree**	node struct that contains two pointers

Alan Turing: Crash Course Computer Science #15

Study online at https://quizlet.com/_6t7u25

1.	**Alan Turing**	English mathematician who conceived of the Turing machine and broke German codes during World War II (1912-1954)
2.	**Entscheidungs Problem or Decision Problem**	This problem asks for an algorithm that takes as input a statement of a first order logic and answers "Yes" or "No" according to whether the statement is universally valid?
3.	**Alonzo Church**	developed a system of mathematical expressions called Lambda Calculus
4.	**Turing Machine**	theoretical computing device equipped with an infinitely long memory tape which stores symbols, and a device called a read-write head which can read and write or modify symbols on that tape
5.	**state variable**	can hold a piece of information about the current state of the machine, and a set of rules that describes what the machine does given a state and the current symbol the head is reading.
6.	**The rule can be to**	write a symbol on the tape, change the state of the machine, move the read-write head to the left or right by one spot, or any combination of these actions
7.	**Turing showed that this simple, hypothetical machine can**	perform any computation if given enough time and memory.
8.	**Turing Complete**	is a term used in computability theory to describe abstract machines, usually called automata. If it can be used to emulate a Turing machine. It is also called computationally universal.
9.	**Halting Problem**	"Is there an algorithm that can determine given a description of a Turing Machine and the input from its tape, whether the machine will run forever or halt?"

10.	**How did Turing reasoned that the Halting Problem is unsolvable?**	If H says the program halts, then we'll make our new machine loop forever. If the answer is no, it doesn't halt, we'll have the new machine output a "no" and halt. In essence, we're building a machine that does the opposite of what H says. Halt if the program doesn't halt, and run forever if the program halts.
11.	**UK's Government Code and Cypher School**	British code-breaking group based out of Bletchley Park
12.	**Enigma Machine**	German secret code machine
13.	**encryption**	Process of converting readable data into unreadable characters to prevent unauthorized access.
14.	**The Enigma Machine behavior was defined by**	a series of reorderable rotors on the top of the , each with 26 possible rotational positions. There was also a plug-board at the front of the machine that allowed pairs of letters to be swapped. In total, there were billions of possible settings.
15.	**The Bombe**	an electro-mechanical device used by British cryptologists to help decipher German Enigma-machine-encrypted secret messages during World War II
16.	**Turing's most famous post-war contribution was to**	artificial intelligence
17.	**Turing Test**	One method of determining the strength of artificial intelligence, in which a human tries to decide if the intelligence at the other end of a text chat is human.
18.	**Turing Award**	The most prestigious technical award given by the Association for Computing Machinery; it is for 'contributions of lasting and major technical important to the computer field'
19.		

	In 1952 burglary at his home revealed	Turing was gay in a time when homosexuality was illegal in the United Kingdom.
20.	**While on probation with hormonal treatment to suppress his sexuality,**	but it altered his mood and personality. Although the exact circumstances will never be known, it's most widely-accepted that Alan Turing took his own life by poison in 1954.

Software Engineering: Crash Course Computer Science #16

Study online at https://quizlet.com/_60biet

1.	**Microsoft Office has roughly**	40 million lines of code
2.	**Software Engineering**	a disciplined approach for constructing information systems through the use of common methods, techniques, or tools
3.	**Margaret Hamilton**	Software engineer who helped NASA prevent serious problems on the Apollo 11 mission to the moon.
4.	**Margaret Hamilton is quoted as saying "It's kind of like a root canal:**	you waited till the end, there are things you could have done beforehand. It's like preventative healthcare, but it's preventative software."
5.	**Breaking big programs into smaller functions allows**	many people to work simultaneously.
6.	**The solution is to package functions into hierarchies, pulling related code together into**	"objects." Then the "parent" object that contains all of the "children" objects.
7.	**Object Oriented Programming**	is the idea of packing up functional units into nested objects
8.	**Application Programming Interface (API)**	A set of software routines that allows one software system to work with another.
9.	**If a function is marked as "private",**	it means only functions inside that object can call it.
10.	**If a function is marked as "public",**	other objects can call it.

11.	**Integrated Development Environment (IDE)**	Provides a developer with a way to create a program, run the program, and debug the program all within one application.
12.	**VIM**	is a clone, with additions, of Bill Joy's vi text editor program for Unix. It was written by Bram Moolenaar based on source for a port of the Stevie editor to the Amiga and first released publicly in 1991
13.	**Don't be the person who**	parachutes a load of uncommented and undocumented code into someone's lap, so they literally have to go line by line to understand what the code is doing
14.	**Good documentation helps programmers when they**	revisit code they haven't seen for awhile, but it's also crucial for programmers who are totally new to it.
15.	**"read-me's"**	standalone files which tell other programmers to read that help file before diving in.
16.	**Comments**	are specially-marked statements that the program knows to ignore when the code is compiled. They exist only to help programmers figure out what's what in the source code.
17.	**Source Control, aka version control or revision control**	A component of software configuration management that in charge of the management of changes to documents, computer programs, large web sites, and other collections of information.
18.	**code repository**	Software development is a collaborative effort and large software projects require teams of developers who may simulta-

	neously work on different parts of the code. Act as a central storage point for developers to place their source code.
19. **committing code**	adds the latest changes to part of the source code to the repository, making these changes part of the head revision
20. **While a piece of code is checked out, and presumably getting updated or modified,**	other programmers leave it alone.
21. **The master version of the code, stored on the server, should always compile without errors and run with minimal bugs. But sometimes bugs do creep in. Fortunately, source control software keeps track of all changes, and**	if a bug is found, the whole code, or just a piece, can be rolled back to an earlier, stable version. It also keeps track of who made each change, so coworkers can send nasty, I mean, helpful and encouraging emails to the offending person.
22. **Quality Assurance (QA)**	a team rigorously tests out a piece of software, attempting to create unforeseen conditions that might trip it up
23. **Beta Software**	An early version of a software program that is often provided free to users to test for bugs before the program is sold to consumers.
24. **Alpha Version**	Is still rough and buggy, so it's only tested internally.

Integrated Circuits & Moore's Law: Crash Course Computer Science #17

Study online at https://quizlet.com/_60bo3b

1.	**discrete components**	Individual electronic parts such as resistors, diodes, capacitors, and transistors.
2.	**vacuum tubes**	They were electronic devices that controlled the flow of electricity in and out of a computer. They looked like long light bulbs.
3.	**Resistors**	Devices that apply resistance to a circuit to manage current
4.	**capacitors**	have the ability to store and discharge electrical potential energy
5.	**diodes**	a device that allows the flow of electrical current in only one direction
6.	**soldered**	To join (metal pieces) with a melted metal compound
7.	**Tyranny of Numbers**	A problem faced in the 1960s by engineers who were unable to increase the performance of their designs due to the huge number of components involved.
8.	**transistors**	invented by Bell Laboratories in 1947, these tiny pieces of technology, which receive and amplify radio signals, make portable radios possible
9.	**Integrated Circuits**	chips that contain large numbers of tiny transistors fabricated into a semiconducting material called silicon
10.	**Germanium (Ge)**	a lustrous, hard, grayish-white metalloid in the carbon group with atomic number 32
11.	**PCB**	Printed Circuit Board are mass manufactured boards that have all the metal wires etched right into them to connect components together.
12.	**Photolithography**	a process to transfer patterns and make layered materials, like a semiconductor, using photosensitive substances & selective light exposure

13.	**semiconductor**	A substance that can conduct electricity under some conditions
14.	**Silicon**	atomic number 14, a hard and brittle crystalline solid with a blue-grey metallic lustre; and it is a tetravalent metalloid and semiconductor
15.	**oxide layer**	layer of an oxygen compound
16.	**photoresist**	a material used in a silicon chip fabrication process that is chemically changed by light, allowing it to be patterned by a mask
17.	**photomask**	an opaque plate with holes or transparencies that allow light to shine through in a defined pattern
18.	**doping**	Process of adding impurities to a semiconductor to increase its conductivity.
19.	**Timing is**	super important in photolithography in order to control things like doping diffusion and etch depth.
20.	**metalization**	process produces a thin-film metal layer that will serve as the required conductor pattern for the interconnection of the various components on the chip
21.	**Bipolar Junction Transistor (BJT)**	A transistor type having two PN junctions, configured as NPN or PNP. May be used as an amplifier or switch.
22.	**Moore's Law**	The observation that the number of transistors in a dense integrated circuit doubles about every two years.
23.	**1968, Robert Noyce and Gordon Moore teamed up and founded a new company,**	combining the words Integrated and Electronics... Intel... the largest chip maker today.
24.		

	very-large-scale integration, or VLSI software	used to automatically generate chip designs using techniques like logic synthesis, where whole, high-level components can be laid down, like a memory cache, the software generates the circuit in the most efficient way possible.
25.	**Two significant issues holding us back from further miniaturization**	We're bumping into limits on how fine we can make features on a photomask and it's resultant wafer due to the wavelengths of light used in photolithography. When transistors get really really small, where electrodes might be separated by only a few dozen atoms, electrons can jump the gap, a phenomenon called quantum tunneling.

Operating Systems: Crash Course Computer Science #18

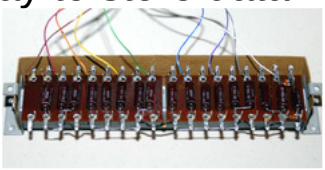
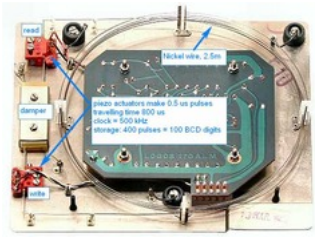
Study online at https://quizlet.com/_60bthp

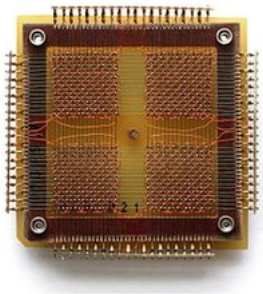
1.	**Operating System (OS)**	System software that provides an interface between the user or application program and the computer hardware.
2.	**batch processing**	Instead of the manual task of loading programs by hand the computers could be given batches. When the computer was done with one, it would automatically and near-instantly start the next.
3.	**peripherals**	An input device, output device, or secondary storage device that is not part of the central processing unit (CPU)
4.	**device driver**	A software program that provides the instructions your computer needs to communicate with a device, such as a scanner.
5.	**I/O**	input and output
6.	**Atlas Supervisor**	The program which managed the allocation of processing resources of Manchester University's Super Computer so that the machine was able to act on many tasks and user programs concurrently.
7.	**The challenge of having many programs running simultaneously on a single computer is that each one is going to need some**	memory, and we can't lose that program's data when we switch to another program. The solution is to allocate each program its own block of memory.
8.	**Virtual Memory**	Space on a hard disk or other storage device that simulates random access memory.

9.	**dynamic memory allocation**	Allocates memory during runtime.
10.	**Memory Protection**	Prevents one process from affecting the confidentiality, integrity, or availability of another.
11.	**Time-sharing**	Each individual user was only allowed to utilize a small fraction of the computer's processor, memory, and so on.
12.	**Unix**	An operating system originally conceived in 1969 by Ken Thompson and Dennis Ritchie of AT&T's Bell Labs.
13.	**kernel**	Unix operating system core functionality of the OS, things like memory management, multitasking, and dealing with I/O
14.	**kernel panic**	A Linux or OS X error from which it cannot recover, similar to a blue screen of death in Windows.
15.	**Microsoft's Disk Operating System, or MS-DOS**	Was just 160 kilobytes, allowing it to fit, as the name suggests, onto a single disk. Though it lacked multitasking and protected memory.
16.	**blue screen of death**	An error condition in which a PC "freezes" and displays a black screen (blue screen prior to Windows Vista); usually turning the computer off and turning it on again clears the error

Memory & Storage: Crash Course Computer Science #19

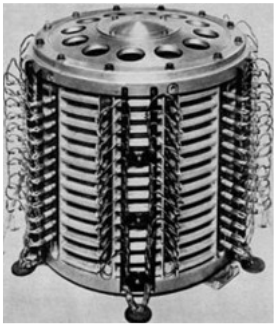
Study online at https://quizlet.com/_60bwe2

1.	**volatile memory**	Storage that is wiped clean when power is cut off from a device
2.	**storage**	the process of maintaining information in memory over time
3.	**Largest program ever punched onto cards was**	stored on 62,500 punch cards.
4.	**Delay Line Memory**	A tube and filled with mercury. At one end put a speaker at the other end a microphone. When you pulse the speaker, it creates a pressure wave. This takes time to propagate to the other end of the tube, where it hits the microphone, converting it back into an electrical signal. And we can use this propagation delay to store data. 
5.	**sequential or cyclic-access memory**	You could only read one bit of data from a tube at any given instant and if you wanted to access a specific bit, like bit 112, you'd have to wait for it to come around in the loop.
6.	**magnetostrictive delay lines**	an electromagnet twisted a long wire one way or the other to represent ones or zeros 
7.	**magnetic core memory**	Memory that stores binary data (0 or 1) in the orientation of magnetic charges in ferrite cores about one-sixeenth-inch in diameter.



8. **Magnetic tape** — a medium for magnetic recording, made of a thin, magnetizable coating on a long, narrow strip of plastic film

9. **Magnetic Drum Memory** — a metal cylinder coated in a magnetic material for recording data. The drum was rotated continuously, and positioned along its length were dozens of read and write heads. These would wait for the right spot to rotate underneath them to read or write a bit of data.



10. **hard disk drive (HDD)** — A direct access storage device used to store and retrieve data from rapidly rotating disks coated with magnetic material.

11. **floppy disk drive (FDD)** — a drive that can hold either a 5½ inch or 3¼ inch floppy disk

12. **floppy disk** — A reusable magnetic storage medium that uses a thin Mylar circular disk as the storage surface.

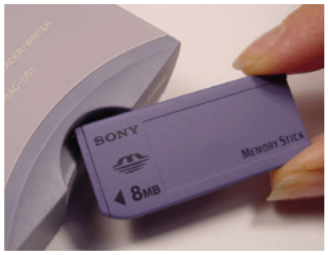
13. **Zip Disks** — Look like floppy disks, but less floppy and holds up to 170 times more data.

14. **laser disc** — 12 inch optical storage device that writes little physical divots in their surface that cause light to be reflected differently, which is captured by an optical sensor, and decoded into 1's and 0's.

15. **USB stick** — A device you can use for saving files from your computer, or moving files between different computers. Sometimes called a flash drive or thumb drive.

16. **Solid State Drive (SSD)** a solid-state storage device that uses integrated circuit assemblies as memory to store data persistently

17. **Stick of Gum drive** Introduced in 1998 with 4MB and 8MB capacities, the original 2" modules looked like a short stick of chewing gum. Somehow left out of the crash course video but forever in the memories of the creator because she had to make a friend return a stupidly expensive one and pay his rent instead.



Files & File Systems: Crash Course Computer Science #20

Study online at https://quizlet.com/_6t7u4m

1.	**file format**	Indicates the manner in which the data a file contains are stored on the disk.
2.	**jpeg**	Most common graphic file format, full color graphic format (16.7 million colors) with a relatively small file size
3.	**.txt**	a computer format that is structured as a sequence of lines of electronic text
4.	**mp3**	A standard format for music files sent over the Internet that compresses music.
5.	**WAV**	(Waveform Audio File Format) an audio coding format standard for storing an audio bitstream of uncompressed audio data.
6.	**metadata**	data about data
7.	**Metadata is stored at the front of the file, ahead of any actual data, in a**	header
8.	**bmp**	is a raster graphics image file format used to store digital images.
9.	**pic**	the native file format for Pictor Paint and Graphics Animation System for Professionals (GRASP) (also by Bridges) and became the first widely accepted DOS imaging standard
10.	**pixel**	Short for "picture element" it is the fundamental unit of a digital image, typically a tiny square or dot which contains a single point of color of a larger image.
11.	**rgb (255,255,0)**	yellow
12.	**rgb (0,0,0)**	black
13.	**rgb (255,255,255)**	white

14. **zip**	a file extension denoting a compressed file, used to save space when sending as attachments
15. **ppt**	a file extension for a presentation file format used by Microsoft PowerPoint
16. **Directory File**	listing of the files stored in memory with their metadata about these files, like when they were created and last modified, who the owner is, and if it can be read, written or both.
17. **Flat File System**	Early information technology systems used this system that is characterized by independent programs and data sets, high degrees of data redundancy, and difficulty in achieving cross functional reporting.
18. **Slack Space**	the space left over from the end of the file to the end of the cluster
19. **Forensic teams can "recover" data from computers that has been deleted because it**	didn't actually erase the file's data in storage, it just deleted the record of it. At some point, that block will be overwritten with new data, but until then, it just sits there.
20. **fragmentation**	Files getting broken up across storage
21. **defrag**	The Windows command that examines a magnetic hard drive for fragmented files and rewrites these files to the drive in contiguous clusters.
22. **Hierarchical File System (HFS)**	The system that store files, consisting of directories and subdirectories that can be nested.
23. **Root Directory**	The top level of the filing structure in a computer system. In Windows computers, the root directory of the hard drive is represented as C:\.

Compression: Crash Course Computer Science #21

Study online at https://quizlet.com/_60c5p5

1.	**Run Length Encoding (RLE)**	A compression algorithm that represents an image in terms of the length of runs of identical pixels
2.	**Lossless Compression**	a data compression algorithm that allows the original data to be perfectly reconstructed from the compressed data
3.	**DFTBA**	Don't forget to be awesome
4.	**Huffman tree**	A tree having the nodes representing symbols as the leaves, with more frequent symbols having nodes progressively closer to the root. The most frequent symbols are the closest to the root node. Note that internal nodes cannot represent symbols. The edges connecting children to parents are either 0 or 1. Convention must be established (e.g., left children have 0s, right children have 1s). The code for a symbol is built from 0s and 1s along the path from the root down to the desired symbol. Codes must be prefix free. abc:7 / \ ab:3 c:4 / \ a:1 b:2
5.	**code dictionary**	The definition of a code is a set of rules or a system of communication, often with randomly assigned numbers and letters given specific meanings.
6.	**Lossy Compression**	(or irreversible compression) a data compression method that uses inexact approximations, discarding some data to represent the content. Most commonly seen in image formats like .jpg.
7.		

	Perceptual Coding	Uses analysis of auditory masking to prioritize elements of a sound as our ears do; saves data by selectively lowering the quality of less-important elements.
8.	**psychophysics**	the study of relationships between the physical characteristics of stimuli, such as their intensity, and our psychological experience of them
9.	**temporal redundancy**	Redundancy that reflects repetition from one image to the next. Video compression techniques exploit different types of redundancy in video.
10.	**MPEG-4**	Format used for Blu-Ray Discs (better quality than MPEG-2), standard for streaming videos over the Internet
11.	**bandwidth**	The amount of data that can be transmitted over a network in a given amount of time.

Keyboards & Command Line Interfaces: Crash Course Computer Science #22

Study online at https://quizlet.com/_60c9mr

1.	**Human-Computer Interaction (HCI)**	The study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.
2.	**keyboard**	An input device that enters data with letters, numbers, symbols, and special function keys.
3.	**QWERTY**	denoting the standard layout on English-language typewriters and keyboards, having q, w, e, r, t, and y as the first keys from the left on the top row of letters.
4.	**switching barriers**	describe any impediment to a customer's changing of suppliers or products
5.	**touch type**	using all your fingers to type without looking at the keyboard
6.	**Teletype Machine**	electro-mechanically-augmented typewriters that could send and receive text over telegraph lines
7.	**Command Line Interface (CLI)**	A graphic-free user interface, such as the Command Prompt program in Windows, where technicians can enter commands more quickly and with more flexibility than in a GUI (graphical user interface) environment.
8.	**CONCATENATE function**	A function that joins two or more text strings into one text string
9.	**Virtual teletype or glass teletype machines became known as**	terminals
10.	**Zork**	PC game; interactive fiction: all text and typing
11.	**MUDS or multi-user dungeons**	Fantasy role-playing games on computers such as "Dungeons and Dragons", they were the predecessors of today's online games with hundreds of thousands of players.

12. **MMORPG** Massively Multiplayer Online Role Playing Game

Crash Course episode 23

Study online at https://quizlet.com/_2ix64r

1.	**what inventions during the civil war helped with the industurial boom that followed?**	the telegraph, clothing, and transcontinental railroads
2.	**what three things help us to understand the rapid authorization that happened after the civil war?**	Geography: large land and resources Law Demography
3.	**what natural resources did the US have at the end of the 1800's**	coal, iron, steal, and wheat
4.	**what was one of the biggest contributors to the increase in the American population at the end of the 1800's?**	immigration because it boosted the economic growth
5.	**examples of how the railroad improved our economy**	time zones, national brands and markets, and also provided a large amount of jobs
6.	**what are two terms to describe the wealthy businessmen at the end of the 1800's?**	captains of industry and robber barons
7.	**who became the wealthiest individual in the world??**	John D. Rockefeller because he owned so much of the oil companies
8.	**Cornelius Vanderbilt**	became rich from his business of shipping
9.	**who were the "unsung" heroes of industrialization**	workers
10.	**what was the name of the first national labor union?**	knights of labor
11.	**what type of issues did the American federation of labor focus on?**	pay, hours, and safety

The Cold War and Consumerism: Crash Course Computer Science #24

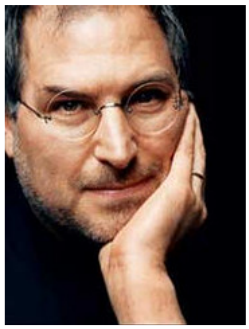
Study online at https://quizlet.com/_60cefl

1.	**Cold War**	the power struggle between the Soviet Union and the United States after World War II contributing massive government spending on science and engineering
2.	**Manhattan Project**	A secret U.S. project for the construction of the atomic bomb.
3.	**UNIVAC I was used by CBS to predict the results of the 1952 US Presidential Election. With just one per cent of the vote**	the computer correctly predicted an Eisenhower landslide while pundits favoured Stevenson.
4.	**MEMEX**	A hypothetical machine proposed by Vannevar Bush that would allow users to store and retrieve information via associative links
5.	**National Science Foundation (NSF)**	An independent agency of the U.S. government that promotes the advancement of science and engineering.
6.	**Apollo Guidance Computer**	a digital computer produced for the Apollo program that was installed on board each Apollo command module and Apollo Lunar Module. The AGC provided computation and electronic interfaces for guidance, navigation, and control of the spacecraft
7.	**video arcade**	a public place where there are a lot of video games that you play by putting money in the machines
8.	**In just three decades computers had evolved from machines where you could literally walk inside of the Su-**	to the point where a child could play with a handheld toy containing a microprocessor many times faster.

	per U, assuming you had government clearance,	
9.	**Government funding like the United States provided during the Cold War enabled**	early adoption of many later computer technologies. This funding helped float entire industries relating to computing long enough for the technology to mature and become commercially feasible.
10.	**Consumerism**	A preoccupation with the purchasing of material goods.

The Personal Computer Revolution: Crash Course Computer Science #25

Study online at https://quizlet.com/_60cqwp

1.	**Microcomputer**	A single-user computer that comes in many forms, such as a desktop model, a portable laptop/notebook, or a handheld computer; also called a personal computer.
2.	**The Altair 8800**	The first commercially successful personal computer
3.	**MITS**	Micro Instrumentation and Telemetry Systems (developed the Altair 8800 microcomputer in 1973)
4.	**BASIC**	Beginner's All-purpose Symbolic Instruction Code is a family of general-purpose, high-level programming languages whose design philosophy emphasizes ease of use.
5.	**interpreter**	Converts a program written in a higher level language into a lower level language and executes it, beginning execution before converting the entire program.
6.	**Homebrew Computer Club (1975-1986)**	An early computer hobbyist group in Silicon Valley which met from March 5, 1975 to December 1986, and was depicted in the films Pirates of Silicon Valley and Jobs, as well as the PBS documentary series, Triumph of the Nerds
7.	**Steve Jobs**	computer industry giant who started Apple Computers, Macintosh, I-Phone, etc. 
8.	**Steve Wozniak**	Co-founder of Apple and creator of the Apple I and Apple II computers

9. **Bill Gates** — American computer software designer who Co-founded Microsoft and built it into one of the Largest computer software manufacturers

10. **Paul Allen** — Co-founder of Microsoft and owner of the NBAs Portland Trailblazers

11. **Apple II** — Apple's earliest product that sold as a complete system that was professionally designed and manufactured. It also offered rudimentary color graphics and sound output, amazing features for a low machine.

12. **TRS-80 Model I**

	a desktop microcomputer launched in 1977 and sold by Tandy Corporation through their Radio Shack stores.
13. **Commodore PET 2001**	the first personal computer sold to the public and formed the basis for their entire 8-bit product line, including the Commodore 64
14. **1977 Trinity**	Byte magazine described Commodore's, Tandy's and Apple's race to own the personal computing market as...
15. **1979s VisiCalc**	the first spreadsheet program - which was infinitely better than paper - and the forbearer of programs like Microsoft Excel and Google Sheets.
16. **dirty dozen**	A crack team of twelve engineers led by engineer Bill Sydnes—which, Lowe promised, could design a prototype in 30 days. The crude prototype barely worked when he demonstrated it in August, but Lowe presented a detailed business plan that proposed that the new computer have an open architecture, use non-proprietary components and software, and be sold through retail stores, all contrary to IBM practice.
17. **IBM PC**	An early, commercially successful personal computer system that featured a 4.77 MHz Intel 8088 processor, 64 KB RAM, and a floppy disk drive. Ancestor to today's PCs.
18. **IBM Compatible**	

		Platform that used Intel processors and DOS or Windows operating systems
19.	**closed architecture**	proprietary designs that typically prevent people from adding new hardware to their computers
20.	**Macintosh**	The brand name of a family of personal computers (hardware) and an operating system (software) from Apple, introduced in 1984, a graphical user interface.

Graphical User Interfaces: Crash Course Computer Science #26

Study online at https://quizlet.com/_60csvn

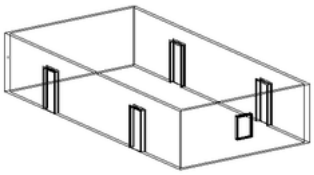
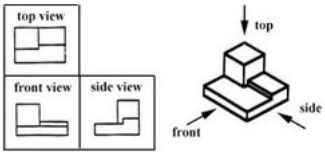
1.	**Instead of having to**	remember the right commands to type in _____**, a graphical user interface sl**
2.	**Graphical User Interface (GUI)**	A visual display on a computer's screen that allows you to interact with your computer more easily by clicking graphical elements.
3.	**Point and click**	"A means of data entry in which the user moves the computer's cursor by way of a mouse, up-and-down arrows, or some other pointing or navigational device to choose an icon or data element"
4.	**intuitive**	knowing something by instinct
5.	**Douglas Engelbert**	Worked with hypertext and had the idea of GUIs. Created the computer mouse.
6.	**Augmenting Human Intellect**	report by Engelbert that stated that the complexity of the problems facing mankind was growing faster than our ability to solve them. Therefore, finding ways to augment our intellect would seem to be both a necessary and a desirable goal.
7.	**1968 Fall Joint Computer Conference**	Featured Englebert's ninety minute demo demonstrating many features of modern computing - bitmap graphics, video conferencing, word processing, and collaborative, real-time editing documents. There were also precursors to modern GUIs, like the mouse and multiple windows, although they couldn't overlap.
8.	**Xerox PARC 'Alto' 1974**	-first working computer Xerox Palo Alto Research Center -built new ideas such as networks, object-oriented programming -thousands of prototypes manufactured -most ideas ended up in Apple ac

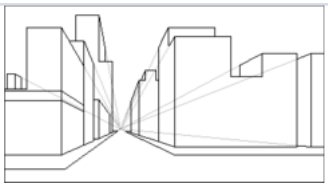
9. **desktop**	the area of the screen in graphical user interfaces against which icons and windows appear
10. **window**	This is what appears when you open a folder, document or program.
11. **WIMP interface**	Stands for: Windows, Icons, Menus & Pointers. It describes a type of user interface where the user selects icons and menu items with a pointer of some kind (with a mouse, stylus or finger).
12. **Widgets**	reusable graphical building blocks, things like buttons, check boxes, sliders, and tabs which were also drawn from real world objects to make them familiar
13. **Event-driven programming**	a programming approach whereby the program's behavior is controlled by writing code that responds to various events that occur, such as Button clicks.
14. **Xerox Star system**	The first commercial system to incorporate various technologies that have since become standard in personal computers, including a bitmapped display, a window-based graphical user interface, icons, folders, mouse (two-button), Ethernet networking, file servers, print servers, and e-mail.
15.	

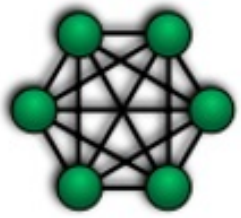
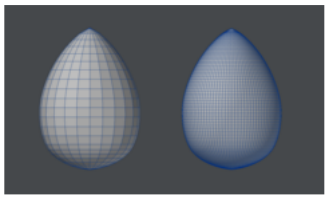
	Where did xerox come up with the terms cut, paste and copy?	You would literally cut text out with scissors and then paste it with glue into the spot you wanted in another document, then you photo-copied the page to flatten it back down into a single layer, making the change invisible.
16.	**moot**	Open to discussion and debate, unresolved
17.	**WYSIWYG**	An acronym that stands for "What you see is what you get." And allows you to see what the final document will look like.
18.	**Xerox approached Apple, hoping to partner with them. Ultimately Xerox was able to**	buy a million dollar stake in Apple before its highly anticipated IPO, but it came with an extra provision: disclose everything cool going on at Xerox Parc.
19.	**Apple Lisa**	Apple's attempt at duplicating Xerox's Alto but it failed and was too expensive (25K)
20.	**Microsoft BOB**	A whole virtual room on your screen with applications embodied as objects that you could put onto tables and shelves. It even came with a crackling fireplace and a virtual dog to offer assistance.

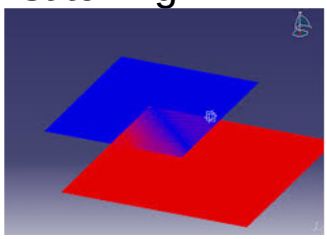
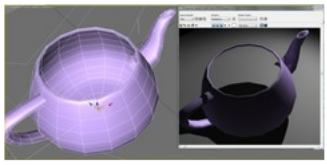
3D Graphics: Crash Course Computer Science #27

Study online at https://quizlet.com/_60cugx

1.	**3D Graphics**	A type of digital graphics format that displays a three-dimensional image on a two-dimensional space.
2.	**zed**	Z
3.	**How many coordinates in a 3d graphic?**	3 - X,Y, Zed
4.	**3D Projection**	Produced by algorithms that are responsible for flattening 3D coordinates onto a 2D plane
5.	**Wireframe modeling**	Computer modeling that represents the form of a three-dimensional object or building by specifying the vertices and edges off all mathematically continuous surfaces, including opposite sides and all internal components normally hidden from view. 
6.	**Orthographic Projection**	A method of representing three-dimensional objects on a plane having only length and breadth. Also referred to as Right Angle Projection. 
7.	**Perspective Projection**	A method of projection in which a three dimensional object is represented by projecting all its points to a picture plane by straight lines converging at an arbitrarily fixed point representing the eye of the owner.

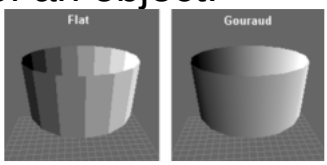


8.	**Mesh**	collection of polygons to create a 3d image 
9.	**The denser the mesh,**	the smoother the curves and the finer the details, But, that also increases the polygon count, which means more work for the computer. 
10.	**Algorithms for simplifying meshes**	Balance model fidelity versus polygon count, because if the count goes too high, the frame rate of an animation drops below what users perceive as smooth.
11.	**Reason triangles are used**	Three points in space unambiguously define a plane, so there is only one answer.
12.	**Scanline rendering**	an algorithm for visible surface determination. Scene elements are projected onto a 2D viewing plane, clipped to the image and rendered by filling in the affected pixels. It works on a row-by-row basis rather than a polygon-by-polygon or pixel-by-pixel basis.
13.	**Anti-aliasing**	Partially fills in pixel edges with additional colors, resulting in smooth-edge type and an increased number of colors in the image.

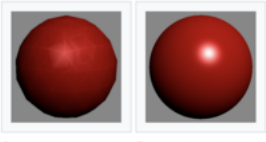
14.	**occlusion**	When only some are visible because some objects are hidden behind other objects in the scene.
15.	**Painter's Algorithm**	A sort algorithm, and arrange all the polygons in the scene from farthest to nearest, then render them in order.
16.	**Z-Buffering**	An algorithm in which the graphics processor stores the Z-axis value of each pixel in a special area of memory called the Z-buffer . Different objects can have the same x- and y-coordinate values, but with different z-coordinate values. The object with the lowest z-coordinate value is in front of the other objects, and therefore that's the one that's displayed.
17.	**Z-Fighting**	a phenomenon in 3D rendering that occurs when two or more primitives have similar or identical values in the z-buffer. Also called "stitching" 
18.	**Back-face culling**	To save processing time, the back side of polygons are often ignored in the rendering pipeline, which cuts the number of polygon faces to consider in half. Except when there's a bug that lets you get inside of those objects and look outwards. Then the avatar head or ground becomes invisible 
19.	**3d shading**	When a light source is added each polygon is going to be illuminated a different amount based on reflection and distance.

20. **flat shading** — an even application of shading

21. **Gouraud shading** — a per-vertex color computation method used in computer graphics to simulate the differing effects of light and colour across the surface of an object.



22. **Phong Shading** — a per-fragment color computation method used in computer graphics to simulate the differing effects of light and colour across the surface of an object.



23. **Texture Mapping** — The application of patterns or images to three-dimensional graphics to enhance the realism of their surfaces. The texturing algorithm will consult the texture in memory, take the average color from the corresponding region, and fill the polygon accordingly.

Computer Networks: Crash Course Computer Science #28

Study online at https://quizlet.com/_60in60

1.	**LAN**	(Local Area Network) A computer network limited to the immediate area usually the same building or floor of a building or a school campus.
2.	**Ethernet**	a system for connecting a number of computer systems to form a local area network, with protocols to control the passing of information and to avoid simultaneous transmission by two or more systems.
3.	**MAC address**	A Media Access Control address is a hardware address that uniquely identifies each node on a network.
4.	**CSMA**	Carrier Sense Multiple Access--A method used by Ethernet networks to allow shared usage of a baseband network and avoid collisions.
5.	**bandwidth**	the sense in multiple access and the rate at which the carrier and transmit data
6.	**collision**	When network traffic increases, the probability that two computers will attempt to write data at the same time also increases. When that happens the data gets all gobbled up, like two people trying to talk on the phone at the same time.
7.	**exponential back-off**	Backing off behavior using an exponentially growing wait time.
8.	**Collision Domain**	The portion of an Ethernet network in which collisions could occur if two nodes transmit data at the same time.
9.	**Network Switch**	network controller which connects network devices and allows effective communication between these devices
10.	**The bigger the networks the more likely that there are**	multiple paths to get data from one location to another.
11.	**Network Router**	

		A device found at each intersection on the Internet backbone that examines the IP address of incoming data, and forwards the data towards its destination.
12.	**circuit switching**	a dedicated connection is formed between two points and the connection remains active for the duration of the transmission
13.	**Message Switching**	A telecommunications methodology that controls traffic in which a complete message is sent to a concentration point and stored until the communications path is established.
14.	**hop count**	The counter associated with a packet as it moves from node to node within a wide area network. Every time the packet moves to the next node, the hop count is increased/decreased by 1.
15.	**Hop Limit**	The number of times that an IPv6 packet can be forwarded by routers on the network; similar to the TTL field in IPv4 packets.
16.	**Internet Protocol (IP)**	A communication standard that enables computers to route communications traffic from one network to another as needed.
17.	**IP address**	A number that uniquely identifies each computer or device connected to the Internet.
18.	**TCP/IP**	Transmission Control Protocol/Internet Protocol. Protocol that connects computers to the Internet. Tells computers how to exchange information over the Internet.
19.	**Packet Switching**	a mode of data transmission in which a message is broken into a number of parts that are sent independently, over whatever route is optimum for each packet, and reassembled at the destination.
20.	**ICMP**	Internet Control Message Protocol A TCP/IP protocol that is used by devices to communicate updates or error information to other devices.

21. **BGP**	Border Gateway Protocol A core routing protocol that bases routing decisions on the network path and rules.
22. **ARPANET**	A computer network developed by the Advanced Research Project Agency (now the Defense Advanced Research Projects Agency) in the 1960s and 1970s as a means of communication between research laboratories and universities. It was the predecessor to the Internet.

The Internet: Crash Course Computer Science #29

Study online at https://quizlet.com/_60ixsi

1.	**internet**	A global network connecting millions of computers, making it possible to exchange information. With a backbone made up of gigantic routers with super high-bandwidth connections running between them
2.	**wifi router**	uses wireless technology to communicate with computing devices over short distances of up to 66 feet
3.	**Wide Area Network (WAN)**	A network that connects devices in geographically separated areas.
4.	**Internet Service Provider (ISP)**	A company that connects computers to the internet for a fee. Abbreviated to ISP.
5.	**Traceroute program**	provides delay measurement from source to router along end-end internet path towards destination
6.	**User Datagram Protocol (UDP)**	Gets the packet to the right program running on that computer and runs checksum, which allows the data to be verified for correctness. It can tell you if the data has been corrupted but can fix it and it cannot tell you if it got to its destination.
7.	**Transmission Control Protocol (TCP)**	provides reliable, ordered, and error-checked delivery of a stream of packets on the internet. TCP is tightly linked with IP and usually seen as TCP/IP in writing.
8.	**ACK**	Acknowledgement
9.	**lag-fragged**	While playing an online game, the game lags and when it's over, you or whoever you were shooting at is dead.
10.	**port number**	The address on a host where an application makes itself available to incoming data.
11.	**Domain Name System (DNS)**	The system administered by the Internet Corporation for Assigned Names (ICANN) that assigns names to each site on the Internet.
12.	**Physical Layer**	

	The lowest, or first, layer of the OSI model. Protocols in this layer generate and detect signals so as to transmit and receive data over a network medium. These protocols also set the data transmission rate and monitor data error rates, but do not provide error correction. "electrical signals on wires, or radio signals transmitted through the air in the case of wireless networks"
13. **Data Link Layer**	The second layer in the OSI model. This layer bridges the networking media with the Network layer. Its primary function is to divide the data it receives from the Network layer into frames that can then be transmitted by the Physical layer. "MAC addresses, collision detection, exponential backoff"
14. **OSI Model**	The seven layers describing network functions
15. **Network Layer**	The third layer in the OSI model. Protocols in this layer translate network addresses into their physical counterparts and decide how to route data from the sender to the receiver. "all the switching and routing technologies that we discussed operate"
16. **Transport Layer**	The fourth layer of the OSI model. In this layer protocols ensure that data are transferred from point A to point B reliably and without errors. this layer services include flow control, acknowledgment, error correction, segmentation, reassembly, and sequencing. "UDP and TCP, which are responsible for point to point data transfer between computers, and also things like error detection and recovery when possible"
17. **Session Layer**	The fifth layer in the OSI model. This layer establishes and maintains communication between two nodes on the network. It can be considered the "traffic cop" for network communications. "TCP and UDP are used to open a connection, pass information back and forth, and then close the connection when finished"

The World Wide Web: Crash Course Computer Science #30

Study online at https://quizlet.com/_60j1n7

1.	**World Wide Web (WWW)**	a huge distributed application running on millions of servers worldwide, accessed using a special program called a web browser
2.	**The world wide web is not interchangeable with the _____.**	internet
3.	**web page**	A document which can display text, graphics, audio, video and other elements through a web browser.
4.	**hyperlink**	A block of text or a graphic that when mouse-clicked takes the user to a new location to an internal or external page.
5.	**Associative Indexing**	whereby any item may be caused at will to select another immediately and automatically
6.	**Uniform Resource Locator (URL)**	Internet address that identifies hypertext documents
7.	**Web Server**	A computer that runs special software to serve up Web content by responding to requests.
8.	**The standard port number for Web servers is _____.**	port 80
9.	**Hypertext Transfer Protocol (HTTP)**	The set of rules that controls the transfer of web pages. It is a protocol that allows web browsers to talk to web servers.
10.	**GET request**	An HTTP method in which the client requests data such as a web page. The client might provide some data as part of the request in the query string of the URL.
11.	**404**	Unknown trouble

12.	**HTTP added status codes which**	prefixed any hypertext that as sent following a GET request.
13.	**200**	"OK, I've got the page, and here it is."
14.	**Hypertext Markup Language (HTML)**	protocol that controls how Web pages are formatted and displayed
15.	**hyperlink reference**	the URL address of a Web page the link "goes to", the target of the link - "href"
16.	**<a>**	Defines a hyperlink
17.	**<ol>**	Defines an ordered list
18.	**<li>**	Defines a list item
19.	**<h1>**	Heading 1
20.	**<h2>**	Heading 2
21.	**If you save HTML text into Notepad or Text Edit and name it something like "test.html", you should be able to open it by**	dragging it into your computer's Web browser.
22.	**Cascading Style Sheets (CSS)**	Allow you to specify styles for various Web page elements.
23.	**CERN**	Conseil Européen pour la Recherche Nucléaire, a provisional body founded in 1952 with the mandate of establishing a world-class fundamental physics research organization in Europe.
24.	**Mosaic Web browser**	Developed at the University of Illinois, at Urbana-Champaign in 1993, it was the first browser that allowed graphics to be embedded alongside text.

25. **Yahoo**	Initially named Jerry and David's Guides to the World Wide Web it was a maintained webpage which served as directories hyper-linking to other websites
26. **search engines**	a program that searches for and identifies items in a database that correspond to keywords or characters specified by the user, used especially for finding particular sites on the World Wide Web.
27. **JumpStation**	First web search engine. This consistent of three pieces of software that worked together: web crawler, ever-enlargening index and search algorithm.
28. **Web Crawler**	a program that pre-visits web pages to build an index
29. **Backlinks**	links from external sources pointing to that webpage
30. **Google Algorithm**	A set of clearly defined, logical steps that create a Google search result from keywords and page rank was developed as part of a research project at Stanford University in 1996 called BackRub
31. **What made Google different (and better)?**	Instead of trusting the content on a webpage, they looked at how other website linked to that page. If it was a spam page with the word "cats" written over and over again, no site would link to it. But if the webpage was an authority on cats, then other sites would likely link to it
32. **Net Neutrality**	the principle that all Internet traffic should be treated equally by Internet Service Providers.

Cybersecurity: Crash Course Computer Science #31

Study online at https://quizlet.com/_60j68e

1.	**Cybersecurity**	the state of being protected against the criminal or unauthorized use of electronic data, or the measures taken to achieve this.
2.	**Secrecy, or confidentiality**	means that only authorized people would be able to access or read specific computer systems and data.
3.	**Integrity**	means that only authorized people should have the ability to use or modify systems and data.
4.	**Availability**	means that authorized people should always have access to their systems and data
5.	**denial of service attack**	hackers overload a website with fake requests to make it slow or unreachable for others
6.	**Threat Model**	A threat model is a diagram and description that tells a story of how an attacker could exploit the vulnerability. This is a narrative approach to the attack that should help guide the mitigation techniques that need to be put in place to protect the system at that point. It can define the security of an application and reduces the number of vulnerabilities. It also has the 2 steps of identifying and prioritizing the vulnerabilities.
7.	**Attack Vector**	The path or means by which an attacker gains access to a computer.
8.	**Security Architect**	responsible for evaluating security technologies; design security aspects of the network topology, access control identity management and other security systems; and establish security policies and security requirements
9.	**'What You Know' authentication**	is based on a knowledge of a secret that should be known only to the real user and the computer. ex. username and password
10.	**Brute Force Attack**	An attack on passwords or encryption that tries every possible password or encryption key.

11.	**bot-net**	is slang term for the computers that have become infected with malicious software and can then be used to launch a DDoS attack.
12.	**An 8 character password with all those symbols, numbers, upper and lower case mixed in has more than**	600 trillion combinations.
13.	**An 8-digit numerical pin only has**	100 million combinations.
14.	**English has about 100 thousand words in use, so putting three together would give you roughly**	one quadrillion possible passwords.
15.	**'What you Have' authentication**	is based on possession of a secret token that only the real user has. An example is a physical key and lock.
16.	**'What You Are' authentication**	is based on you. You authenticate by presenting yourself to the computer. Biometric authenticators like fingerprint readers and iris scanners are classic examples.
17.	**Researchers recently showed it's possible to forge your iris just by**	capturing a photo of you.
18.	**Two-factor or Multi-factor Authentication**	using two or more forms of authentication for important accounts

19.	**Access Control**	A security measure that defines who can access a computer, device, or network, when they can access it, and what actions they can take while accessing it.
20.	**Permissions or Access Control Lists**	describe what access each user has for every file, folder and program on a computer.
21.	**Read Permission**	allows a user to see the contents of a file
22.	**Write Permission**	allows a user to modify the content
23.	**Execute Permission**	allows a user to run a file like a program
24.	**Bell-LaPadula Model**	-DoD created this in the 1970's -no 'read up', no 'write down' -first mathematical model of a multilevel sec policy
25.	**Chinese Wall Model**	"Model designed to avoid conflicts of interest by prohibiting one person, such as a consultant, from accessing multiple conflict of interest (Col) categories."
26.	**Biba Model**	An access control model used to ensure integrity. It uses two primary rules: no read down and no write up.
27.	**Malware**	software designed to infiltrate or damage a computer system without the user's informed consent
28.	**implementation bug**	means that the code does not match the design
29.	**security kernel**	a minimal set of operating system software that's close to provably secure
30.	**Independent Verification and Validation**	the code audited by a crowd of security-minded developers
31.	**Defcon**	

	conference in Las Vegas where like-minded hackers and security experts can mingle and share ideas
32. **sandbox application**	a security mechanism for separating running programs, usually in an effort to mitigate system failures or software vulnerabilities from spreading
33. **Virtual Machine**	Essentially simulated computers that live in their own sandbox. If a program goes awry, worst case is it crashes or compromises only the virtual machine on which it's running.

Hackers & Cyber Attacks: Crash Course Computer Science #32

Study online at https://quizlet.com/_60jbbe

1.	**cyber attack**	a malicious and deliberate attempt by an individual or organization to breach the information system of another individual or organization
2.	**hacker**	people that make items/systems do things that they weren't intended to do
3.	**White Hats**	Also known as ethical hackers, these are computer security experts who specialize in penetration testing and other testing methods to ensure that a company's information systems are secure.
4.	**Black Hats**	Hackers who specialize in unauthorized penetration of information systems. They attack systems for profit, fun, or political motivation or as part of a social cause. These penetration attacks often involve modifying and destroying data.
5.	**hacktivist**	a computer hacker whose activity is aimed at promoting a social or political cause.
6.	**social engineering hack**	where a person is manipulated into divulging confidential information or configuring a computer system so that it permits entry by attackers
7.	**phishing**	the fraudulent practice of sending emails purporting to be from reputable companies in order to induce individuals to reveal personal information, such as passwords and credit card numbers.
8.	**Pre-texting**	Creating an invented scenario (the pretext) to convince someone to divulge information.
9.	**Trojan Horses**	programs that masquerade as harmless attachments like a photo or invoice, but actually contain malicious software called malware
10.	**Ransomware**	a type of malicious software designed to block access to a computer system until a sum of money is paid.

11.	**NAND Mirroring**	Where if you have physical access to the computer, you can attach wires to the device's memory chip and make a perfect copy of its contents. You can try a series of passwords until the device starts making you wait. When this happens, you just refresh the memory with the original copy you made, essentially resetting it, allowing you to try more passwords immediately with no waiting
12.	**exploit**	successfully utilizing a bug to gain capabilities or access
13.	**buffer overflow attack**	Inputting so much data that the input buffer overflows. The overflow contains code that takes control of the computer.
14.	**bounds checking**	Makes sure that the index is within the range of the array being referenced. If the index is not valid, an ArrayIndexOutOfBoundsException is thrown. Automatically performed by the index operator.
15.	**Canaries**	unused space after buffers and keep an eye on all those values to see if they change. If they do, they know an attacker is monkeying around with memory. These regions are named after the small birds miners used to take underground to warn them of dangerous conditions.
16.	**code injection attack**	Exploitation of a computer bug that is caused by processing invalid data. The attacker introduces (or "injects") code into a vulnerable computer program and change the course of execution.
17.	**zero-day vulnerability**	software vulnerability that has been previously unreported and for which no patch yet exists
18.	**Worms**	destructive programs that replicate themselves without requiring another program to provide a safe environment for replication
19.	**Why should we do our computer updates?**	A lot of those downloads are security patches.

20.	**DDoS (Distributed Denial of Service)**	is where all the computers in the botnet send a flood of dummy messages. This can knock services offline, either to force owners to pay a ransom of just to be evil.
21.	**Cyber attacks cost the**	global economy roughly half a trillion dollars annually, and that figure will only increase as we become more reliant on computing systems
22.	**Many experts predict that the next war will be fought in**	cyberspace, where nations are brought to their knees not by physical attack, but rather crippled economically and infra-structurally through cyber warfare.
23.	**cracker**	A criminal hacker who uses his computer skills to gain access to computer systems without permission and/or tampers with programs and data, often causing damage or destruction.
24.	**Botnet**	a group of compromised computers or mobile devices connected to a network

Cryptography: Crash Course Computer Science #33

Study online at https://quizlet.com/_60je59

1.	**Defense in Depth**	A defense that uses multiple types of security devices to protect a network. Also called layered security.
2.	**Cryptography**	the art of protecting information by transforming it into an unreadable format, called cipher text
3.	**ciphertext**	A string of text that has been converted to a secure form using encryption.
4.	**Encryption**	Process of converting readable data into unreadable characters to prevent unauthorized access.
5.	**Decryption**	a process that reverses encryption, taking a secret message and reproducing the original plain text
6.	**Ceasar Cipher**	A substitution cipher that shifts characters 3 positions in the alphabet
7.	**key**	a piece of information (a parameter) that determines the functional output of a cryptographic algorithm
8.	**substitution ciphers**	one letter is used for another.
9.	**Substitution ciphers are broken with**	frequency analysis. In English, "e" and "t" are the most frequently occurring letters. You figure out where they are.
10.	**frequency analysis**	a count of the number of cases that fall into each category when the categories are based on one variable
11.	**Permutation Cipher**	a method of encryption by which the positions held by units of plaintext are shifted according to a regular system, so that the ciphertext constitutes a permutation of the plaintext.
12.	**Columnar Transposition Cipher**	writing the plain text out in rows, and then reading the ciphertext off in columns
13.		

	Data Encryption Standard (DES)	A symmetric block cipher that uses a 56-bit key and encrypts data in 64-bit blocks.
14.	**Advanced Encryption Standard (AES)**	A block cipher that uses a 128-bit block size and a 128-, 192-, or 256-bit key size. Practically uncrackable.
15.	**Diffie-Hellman key exchange**	Invented in the 1970s, it was the first practical method for establishing a shared secret key over an unprotected communications channel.
16.	**modular exponentiation**	Applying exponentiation to a value relative to a modulus
17.	**asymmetric encryption**	used in public key encryption, it is scheme in which the key to encrypt data is different from the key to decrypt.
18.	**RSA**	Rivest-Shamir-Adleman is the most common internet encryption and authentication system. The system used an algorithm that involves multiplying two large prime numbers to generate a public key, used to encrypt data and decrypt an authentication, and a private key, used to decrypt the data and encrypt an authentication.
19.	**Block Cipher**	An encryption algorithm in which data is encrypted in "chunks" of a certain length at a time. Popular in wired networks.

Machine Learning & Artificial Intelligence: Crash Course Computer Scienc #34

Study online at https://quizlet.com/_60q26p

1.	**Machine Learning**	the extraction of knowledge from data based on algorithms created from training data
2.	**Artificial Intelligence**	A scientific field that focuses on creating machines capable of performing activities that require intelligence when they are done by people.
3.	**classifier algorithm**	decision process is called classification example: deciding if a moth is a luna moth or an emperor moth
4.	**Features**	are values that usefully characterize the things we wish to classify
5.	**Labeled data**	a group of samples that have been tagged with one or more labels
6.	**scatter plot**	A graph with points plotted to show a possible relationship between two sets of data.
7.	**What do machine learning algorithms do?**	Find optimal separations.
8.	**Decision Boundaries**	The region of a problem space in which the output label of a classifier is ambiguous.
9.	**Confusion Matrix**	a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known
10.	**misclassify**	to assign (someone or something) to an incorrect group or category
11.	**forest**	algorithms that use many decision trees working together to make a prediction
12.	**Support Vector Machine**	Supervised learning classification tool that seeks a dividing hyperplane for any number of dimensions can be used for regression or classification

13.	**Statistics**	Collection of methods for planning experiments, obtaining data, organizing, summarizing, presenting, analyzing, interpreting, and drawing conclusions based on data.
14.	**Artificial Neural Networks (ANNs)**	computer systems that are intended to mimic human cognitive functioning
15.	**Deep Learning**	is part of a broader family of machine learning methods based on learning data representations, as opposed to task-specific algorithms
16.	**weak AI or narrow AI**	is only intelligent at specific tasks
17.	**strong AI**	Artificial intelligence with a generalized learning ability, on par with human intelligence.
18.	**IBM's Watson**	Supercomputer that knows almost anything, but struggles with real world messy data Competed on Jeopardy
19.	**Google's AlphaGo**	a computer program that plays the board game Go
20.	**reinforcement learning**	a type of Machine Learning, and thereby also a branch of Artificial Intelligence. It allows machines and software agents to automatically determine the ideal behaviour within a specific context, in order to maximize its performance.

Computer Vision: Crash Course Computer Science #35

Study online at https://quizlet.com/_60qpz1

1.	**computer vision**	Techniques to let computers and robots see and understand the world around them.
2.	**kernel or filter**	mathematical equation to determine the bigger the color difference between these two sets of pixels, the more likely the pixel is on an edge. If the color difference is small, it's probably not an edge at all.
3.	**Grayscale**	image is one in which the value of each pixel is a single sample representing only an amount of light
4.	**convolution**	operation of applying a kernel to a patch of pixels
5.	**Prewitt Operators**	is used in image processing, particularly within edge detection algorithms. Technically, it is a discrete differentiation operator, computing an approximation of the gradient of the image intensity function.
6.	**Viola-Jones Face Detection**	In order to be detected, the entire face must point towards the camera and should not be tilted to either side. After the detection step is most often followed by a recognition step.
7.	**Convolutional Neural Networks**	Type of feed-forward artificial neural network in which the connectivity pattern between its neurons is inspired by the organization of the animal visual cortex, whose individual neurons are arranged in such a way that they respond to overlapping regions tiling the visual field.
8.	**biometric data**	digital scanning of the physiological or behavioral characteristics of individuals as a means of identification
9.	**CCTV cameras (closed-circuit television)**	a system that allows someone to watch on a television screen what is happening in different parts of a building, town
10.	**Intelligent Tutoring Systems (ITS)**	instructional systems that use artificial intelligence
11.		to take pictures is not the same as to see."

	Computer vision professor Fei-Fei Li recently said, "Just like to hear is not to listen,	
12.	**fidelity**	hi-fi is most commonly a term for the high-quality reproduction of sound or images
13.	**color marker**	able to search pixel by pixel because colors are stored inside of single pixels
14.	**drone**	An unmanned aerial vehicle is an aircraft without a human pilot aboard

Natural Language Processing: Crash Course Computer Science #36

Study online at https://quizlet.com/_60ri5k

1.	**natural language**	Any one of the languages that people speak that evolved naturally.
2.	**natural language processing**	Techniques for processing human languages to enable a computer to understand their meaning.
3.	**nine fundamental types of English words:**	nouns, pronouns, articles, verbs, adjectives, adverbs, prepositions, conjunctions, and interjections
4.	**grammar**	a system of rules that enables us to communicate with and understand others
5.	**Parse Tree**	A structure that starts from the individual tokens in a statement and shows how these tokens can be grouped into predefined grammatical categories.
6.	**codify**	to arrange or set down in writing
7.	**semantic**	Relating to the different meanings of words or other symbols
8.	**Knowledge Graph**	An infrastructure developed by Google to display related information about people, places, and objects.
9.	**chatbot**	A program designed to carry on a conversation with a human user
10.	**ELIZA**	An early natural language processing computer program created from 1964 to 1966 at the MIT Artificial Intelligence Laboratory by Joseph Weizenbaum
11.	**AUDREY**	The first true speech recognition device, in 1952. The machine understood digits 0-9 if speakers paused in between, though it would have to adapt to each user before it could capture their speech with reasonable accuracy.
12.	**HARPY**	1971, DARPA provided funding for a five-year speech recognition project at Carnegie Mellon University. It had

		a 1,011-word vocabulary and could also understand entire phrases, including where different words started and stopped.
13.	**SHOEBOX**	A voice-activated calculator developed by an IBM engineer for the World's Fair in Seattle: It understood 10 digits and six control words — plus, minus, total, subtotal, false, and off — and instructed an attached adding machine to calculate and print out answers to basic spoken math problems.
14.	**What do the speech recognition systems with the best accuracy use?**	deep neural networks
15.	**acoustic signal**	The pattern of frequencies and intensities of the sound stimulus.
16.	**waveforms**	graphic representation of the shape, direction, amplitude, and direction of a particular sound current
17.	**spectogram**	A pattern for sound analysis that provides a three-dimensional display plotting time on the horizontal axis, frequency on the vertical axis, and intensity in color
18.	**Fast Fourier Transform (FFT)**	a mathematical process used for analyzing and processing the Doppler signal to produce the spectral waveform
19.	**formants**	resonant frequencies of the vocal tract
20.	**Phonemes**	in language, the smallest distinctive sound unit
21.	**How many phonemes are there in the English language?**	Roughly 44
22.	**speech synthesis**	

The process by which computers produce sound that resembles spoken words

Robots: Crash Course Computer Science #37

Study online at https://quizlet.com/_60rqov

1.	**humanoid**	having an appearance or character resembling that of a human.
2.	**robot**	A robot is a machine that performs complicated tasks and is guided by automatic controls.
3.	**bots or agents**	A program that can do things without the user of the computer having to give it instructions. Many bots are malware as they are installed without people's permission and can be controlled over the internet and used to send spam or steal data. Also known as web robots.
4.	**Etymology of "robot"**	It was first used in a 1920 Czech play to denote artificial, humanoid characters. The word was derived from "robota", the slavic-language word for a forced laborer, indicating peasants in compulsory service in feudal, nineteenth century Europe
5.	**automatons**	non-electric machine that performed functions automatically
6.	**Mechanical Turk**	It was a chess-playing machine constructed in the late 18th century. Only it was a hoax - there was a dainty human stuffed inside the machine.
7.	**Computer Numerical Control (CNC)**	A numerical control method in which one computer is linked with one machine tool to perform NC functions.
8.	**Unimate Robot**	Was used by General Motors in the 1950's to assemble cars at one of their plants in New Jersey.
9.	**negative feedback loop**	A feedback loop that causes a system to change in the opposite direction from which it is moving
10.	**error**	measurement of how far we are from where we want to be
11.	**proportional-integral-derivative controller**	a control loop feedback mechanism to control process variables and are the most accurate and stable controllers

12.	**proportional value**	which is the difference between the desired value and the actual value at the most recent instant in time or the present
13.	**integral value**	the sum of the error over a window of time
14.	**derivative value**	the rate of change between the desired and actual values
15.	**anticipatory control**	Alteration or adjustment of the motor program even before any interaction with the environment occurs
16.	**What tasks are devilishly difficult for robots?**	walking on two legs, opening a door, picking up objects without crushing them, putting on a t-shirt, or petting a dog
17.	**Lethal autonomous weapons systems (LAWS)**	The formal name for "military robots," military weapons that can function without the continuous involvement of a human operator
18.	**Isaac Azimov**	American writer and professor of biochemistry at Boston University.
19.	**fictional Three Laws of Robotics**	1. A robot may not injure a human being or, through inaction, allow a human being to come to harm. 2. A robot must obey orders given it by human beings except where such orders would conflict with the First Law. 3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Psychology of Computing: Crash Course Computer Science #38

Study online at https://quizlet.com/_60s4vs

1.	**human cognition**	the development of a person's ability to think and understand
2.	**Humans can read, remember, and process information more effectively when**	items are put together in small, meaningful groups.
3.	**Humans can generally juggle 7 items plus or minus 2 in**	short term memory.
4.	**Affordances**	the possibilities for action offered by objects and situations
5.	**knurling**	produces a regularly shaped, roughened surface on a work piece
6.	**Recognition versus recall**	refers to our ability to "recognize" an event or piece of information as being familiar versus designates the retrieval of related details from memory
7.	**Human memory is much better when it's**	triggered by a sensory cue like a word, picture, or sound.
8.	**expertise**	specialized skill or technical knowledge; know-how; expertness
9.	**Good interfaces should offer**	multiple paths to accomplish goals
10.	**Affect-aware systems**	use sensors, sometimes worn, that capture things like speech and video of the face, as well as biometrics like sweatiness and heart-rate.
11.	**Computer-Mediated**	the exchange of messages carried through an intervening system of digital electronic storage and transmitted between two or more people

	Communication (CMC)	
12.	**Emoticons**	Typed symbols that convey emotional aspects of an online message
13.	**Augmented Gaze**	computer vision and graphics software that can warp the head and eyes, making it appear as though the instructor is looking into the camera, right at the remote viewer.
14.	**anthropomorphizing**	to give a nonhuman thing a human form, characteristic or behavior
15.	**Human-robot interaction or HRI**	A field dedicated to studying these interactions, like how people perceive different robot behaviors and forms or how robots can interpret human social cues to blend in and not be super awkward.
16.	**Uncanny Valley**	Used in reference to the phenomenon whereby a computer-generated figure or humanoid robot bearing a near-identical resemblance to a human being arouses a sense of unease or revulsion in the person viewing it.

Educational Technology: Crash Course Computer Science #39

Study online at https://quizlet.com/_60sdgj

1.	**Dancing Plague of 1518**	a case of dancing mania that occurred in Strasbourg, Alsace, in the Holy Roman Empire in July 1518. Around 400 people took to dancing for days without rest and, over the period of about one month, some of those affected collapsed or even died of heart attack, stroke, or exhaustion
2.	**proper toilet paper roll orientation**	the 1891 patent for the toilet paper roll literally states that the end of the roll should be hanging off the exterior, so there's that
3.	**What can you do while watching an educational video to increase what you learn and retain by a factor of 10?**	-make sure the pacing is right for you by using the video speed control -stop the video at the difficult parts and ask yourself questions about what you've watched -try any examples or exercises that are presented in the video on your own
4.	**Massive Open Online Courses**	Online learning designed to enroll large numbers of learners who have access to the Internet and composed of interactive coursework including video lectures, discussion groups, wikis, and assessment quizzes.
5.	**production rules**	"If ... then" or condition-action rules in which the action is carried out whenever the appropriate condition is present
6.	**Domain Model**	a formal representation of the knowledge, procedures, and skills of a particular discipline
7.	**student model**	tracks what production rules a student has mastered, and where they still need practice
8.	**Bayesian knowledge tracing**	an algorithm that models student knowledge in a Hidden Markov Model as a latent variable, updated by observing the correctness of each student's interaction in which they apply the skill in question. It assumes that student knowledge is represented as a set of binary variables, one per skill, where the skill is either mastered by the student or not.

9.	**probability of transit**	probability that the student started off the problem not knowing how to do the step, but learned how to do it as a result of working through the problem
10.	**probability of slip**	possibility that she knows it, but made a careless error or other slip-up
11.	**probability of guess**	possibility that the student got it correct by accident, and doesn't actually understand how to solve the problem
12.	**probability of knowing the skill**	probability that a student has learned how to do a particular skill
13.	**adaptive sequencing**	The capacity of tailoring the order in which learning units are presented to learners to their specific characteristics.
14.	**Educational Data Mining**	digging for patterns in data collected from digital learning/educational environments
15.	**pedagogical**	Characteristic of teaching or teachers

The Singularity, Skynet, and the Future of Computing: Crash Course Computer Science #40

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1.	**complexity brake**	idea that progress is going to level off and be more of an S curve than an exponential one, where as complexity increases, it becomes more difficult to make additional progress
2.	**technological unemployment**	where workers in many job sectors are rendered obsolete by computers, like AIs and robots, that can do their work better and for less pay
3.	**non-routine cognitive work**	requires abstract reasoning, systems thinking, collaboration, and the ability to experiment ex. professions like teachers and artists, novelists and lawyers, and doctors and scientists
4.	**Transhumanist**	an individual who is in favor of human genetic engineering. Believes in the creation of a posthuman, extension of human health-span, eradication of disease, elimination of unnecessary suffering, and augmentation of human intellectual, physical, and emotional capacities.
5.	**Cyborg**	a fictional or hypothetical person whose physical abilities are extended beyond normal human limitations by mechanical elements built into the body.
6.	**digital ascension**	the hypothetical concept of storing (or transferring) a person's personality in more durable media, i.e., a computer
7.	**paradigm shift**	a fundamental change in approach or underlying assumptions.
8.	**Cryptocurrencies**	A digital currency in which encryption techniques are used to regulate the generation of units of currency and verify the transfer of funds, operating independently of a central bank. It solves the problem of counterfeiting (fake replicates of real products) and double spending (The risk that a digital currency can be spent twice).
9.	**3D printing**	builds—layer by layer in an additive process—a three-dimensional solid object from a digital model

10.	**Quantum computing**	uses the principles of quantum physics to represent data and perform operations on these data
11.	**bioinformatics**	application of mathematics and computer science to store, retrieve, and analyze biological data
12.	**dystopian**	relating to or denoting an imagined place or state in which everything is unpleasant or bad, typically a totalitarian or environmentally degraded one.
13.	**Google Glass**	Google's project program for developing a line of hands-free, head-mounted intelligent devices that can be worn as eyewear.